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MODULE 5 HANDBOOK

# Module 5: Data Extensions and Data Management

5 sections, 39 topics explained, plus the detailed lessons, examples, exercises and official links.

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Edition: November 2026. Practice labs marked "account" need a provisioned Marketing Cloud account. Other labs use sample files or paper.

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# Module 5: Data Extensions and Data Management

**Level:** Beginner. Data extensions: how they are created, imported, updated and kept clean.

## Contents


1. 5.1 Data Extension Fundamentals
2. 5.2 Data Extension Configuration
3. 5.3 Data Operations
4. 5.4 Data Management
5. 5.5 Data Sources


## Detailed lessons

Module 5

### 5.1 Creating and configuring data extensions

#### WHAT IS IT?

> A data extension is an SFMC table. Types: standard, sendable (linked to a subscriber key and used for sends), non-sendable, and test. Each has fields with types, a primary key, nullable settings and retention settings.

#### WHY IS IT IMPORTANT?

> Choosing the right type and key decides whether the data can be used for sending and how long it stays.

#### HOW DOES IT WORK AND THE STEPS

1. Create a data extension from the Email Studio or Contact Builder navigation.
2. Add fields with the right type (Text, Number, Date, Email, Phone, Boolean).
3. Set the primary key and make key fields non-nullable.
4. Set retention, for example delete all records after 365 days.

#### REAL-WORLD EXAMPLE

> A sendable data extension for customers with SubscriberKey as the send relationship field.

#### COMMON ERRORS

Choosing text for dates, so date filters fail.

#### INTERVIEW QUESTION

What is a sendable data extension?

#### HANDS-ON EXERCISE

Create a customers data extension with five fields and a primary key in a sandbox account (or write the design if you have no account).

#### OFFICIAL DOCUMENTATION AND FURTHER READING

- Salesforce Help: Marketing Cloud Engagement

Module 5

### 5.2 Importing, updating and cleaning data

#### WHAT IS IT?

> Data can be imported from CSV files, SFTP, Salesforce CRM, APIs and databases. Import options include overwrite, append, and update or insert on the primary key. Data quality checks catch duplicates, bad dates and missing keys.

#### WHY IS IT IMPORTANT?

> Bad imports send emails to the wrong people. Validate before loading.

#### HOW DOES IT WORK AND THE STEPS

1. Prepare the CSV with a header row matching the field names.
2. Use the import activity with the update-or-insert option on the primary key.
3. Check row counts before and after the import.
4. Remove duplicates with a SQL query before the send.

#### REAL-WORLD EXAMPLE

> Import 10,000 rows; the import reports 9,980 rows, so 20 rows had duplicate keys.

#### CODE EXAMPLE, WITH LINE-BY-LINE NOTES

```
-- Find duplicate keys before a send
SELECT SubscriberKey, COUNT(*) AS Copies
FROM Customers
GROUP BY SubscriberKey
HAVING COUNT(*) > 1;
```

#### COMMON ERRORS

Choosing overwrite when append was intended, and losing history.

#### INTERVIEW QUESTION

How would you import daily customer updates without creating duplicates?

#### HANDS-ON EXERCISE

Import a sample CSV with planted duplicates and fix them with an import option and a SQL check.

#### OFFICIAL DOCUMENTATION AND FURTHER READING

- Salesforce Help: import data

## 5.1 Data Extension Fundamentals: every topic explained

**What is a data extension?.** A table in SFMC with defined fields, used to store data and audiences.

**Standard data extensions.** A regular data extension with fields you define.

**Sendable data extensions.** A data extension with a send relationship to subscriber key, so it can be used as a send audience.

**Non-sendable data extensions.** A data extension used for lookups and reference data, not for sending.

**Test data extensions.** A data extension marked for tests, so test sends use it.

**Shared data extensions.** Data extensions shared across business units from the parent.

**Data extension templates.** Predefined structures that create data extensions with required fields.

## 5.2 Data Extension Configuration: every topic explained

**Creating data extensions.** Create from Email Studio or Contact Builder, give a name, external key and fields.

**Field configuration.** Each field has a name, type, length, primary key flag, nullable flag and default value.

**Field types.** Text, Number, Decimal, Date, Boolean, EmailAddress, Phone and Locale.

**Primary keys.** The field(s) that make each row unique and drive update-or-insert imports.

**Nullable fields.** Whether a field can be empty. Key fields must not be nullable.

**Default values.** A value used when none is provided, such as today's date.

**Send relationships.** Links a sendable data extension field to the subscriber key.

**Retention settings.** Rules to delete rows or the whole data extension after a period.

**Data extension permissions.** Who can read, edit or delete the data extension, by role and folder.

## 5.3 Data Operations: every topic explained

**Importing data.** Loading data from files or connectors into a data extension.

**Exporting data.** Writing data extension rows to a file with a Data Extract activity.

**Updating records.** Changing existing rows by key.

**Overwriting records.** Replacing all rows with the new file.

**Appending records.** Adding rows without checking for existing keys.

**Updating and inserting records.** Updating rows that exist and inserting new ones, by primary key.

**Import activities.** Automation Studio activities that import files into data extensions on a schedule.

**File transfer activities.** Moving files between the SFTP and the safehouse, and unzipping or decrypting them.

## 5.4 Data Management: every topic explained

**Data retention.** Keeping data only as long as needed, set per data extension.

**Data deletion.** Removing rows or whole tables as policy or requests require.

**Duplicate handling.** Finding and removing duplicate keys before sending.

**Data validation.** Checking formats, ranges and required fields on import.

**Data cleansing.** Fixing or removing bad values.

**Data standardisation.** Making formats consistent: case, phone format, date format.

**Data quality.** Overall accuracy, completeness and consistency of the data.

**Data governance.** Ownership, rules and audits for marketing data.

## 5.5 Data Sources: every topic explained

**CSV.** Comma-separated files, the most common import format.

**TXT.** Plain text files with other delimiters, such as tab or pipe.

**SFTP.** The secure FTP server in the account used to exchange files.

**Salesforce CRM.** Data synchronised from Sales or Service Cloud through the connector.

**API integrations.** Data pushed or pulled with the REST or SOAP APIs.

**External databases.** Data exported from other systems into files or via middleware.

**Cloud storage.** Storage such as cloud buckets reached through file transfer or integration tools.